Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
1	(a)			(236.052595 – 235.866563) u = (1) i.e. attempt at LHS-RHS 0.186032 [u] can be implied (1) $\times$ 931 (1) Energy released = 173.20 M[eV] (1)	1	1 1 1		4	4	
	(b)			Number of nuclei in 1.2 kg i.e. number method $= \frac{6.02\times10^{23}}{235\times10^{-3}}$ x 1.2 = 3.074×10^{24} **or** $\frac{1.2}{(235 \text{ x } 1.66 \text{ x } 10^{-27})}$ (1) Energy released by 1.2 kg i.e. 1(a) **ecf** $\times$ number $= (3.074\times10^{24})(173.20) = 5.32\times10^{26}$ MeV (1) Conversion and answer i.e. $= (5.32\times10^{26})$ x $(1.6\times10^{-13}) = 8.52\times10^{13}$ (1) **Alternative:** $\frac{0.186032}{235}$ x 1.2 (1) $\times c^2$ (1) = 8.56 $\times 10^{13}$ (1)		3		3	3	
	(c)			Use of 3600 factor e.g. Energy required for 1 light bulb for 1 hour $= 11\times60\times60$ $= 39600$ J (1) Correct use of energy, power equation e.g. Number of bulbs $= \frac{8.52\times10^{13}}{39600}$ [method, ecf for energy 1] $= 2.15\times10^{9}$ so her claim is correct (1) Accept other numbers e.g. 7.92 $\times 10^{13}$ [J] 11.8 [W], 3863 or 3868 [s], 2.36 $\times 10^{10}$ [W]			3	3	2	
				Question 1 total	**1**	**6**	**3**	**10**	**9**	**0**